FLEET TEAM DIGEST

737MAX-FTD-33-19001

Issue Title : Landing and Runway Turnoff Light LED Failures

Airplane Model	ATA	Minor Model(s)
737MAX	3345-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
03/29/2019	08/28/2026	08/21/2026

Estimated Completion Date	Next Update Date
12/31/2027	11/28/2026

Revision Description

No updates for this revision.

Applicability

All 737MAX airplanes.

Description

The LED Landing Light Assembies, P/N 72500410-1/-2/-3/-4, and the Runway Turnoff Light (RTO) Assemblies, P/N 72500415-1/-2, have experienced an increase of field returns. Most of the light assemblies that have been returned from the field exhibit discolored (brown) LEDs. In addition, the silicon Room Temperature Vulcanizing (RTV) adhesive that is applied to the circuit board located near the LEDs also exhibits discoloring (brown). LED failures have also been found to be caused by solder connections cracking on the circuit board.

Boeing has also received reports of failures on the LTRTL Power Supply P/N 72605363-1, which leads to inoperative Landing and Runway Turnoff lights on the respective side of the airplane. There are two LTRTL power supply assemblies per airplane, installed in each wing strakelet. Boeing has recently received multiple reports of power supply failures on recently delivered airplanes.

Status

- It has been determined that the discoloring occurs due to chemical outgassing of polysulfide sealant when present in a confined environment. The LED operational temperature also contributes to the failure mode. Sulfur outgassing of the polysulfide sealant used in the wing root structure is contributing to the field failures of the LED light assemblies.
- The solder connection failures were found to be caused by a Coefficient of Thermal Expansion (CTE) mismatch between the circuit board core material and the solder.
- Honeywell evaluated alternate LEDs that have improved material and production processes. LED operational testing in a polysulfide atmosphere has been completed and a replacement LED has been selected. As a result of testing, it was determined that Osram LEDs provide products that can withstand the exposure to sulfur outgassing. A new circuit board

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layout is required for the replacement Osram LEDs. The new board will also address issues of solder cracking by incorporating a board material change.
- Honeywell is redesigned the RTO assembly and the Main Array assembly of the LTRTL system to address both discolored LED and solder joint failure modes.This was considered a major design change and required full qualification testing. Qualification Testing was successfully completed in May 2023. The new RTO and Landing Light Arrays were incorporated into 737 MAX production starting at Line Number 8900, delivered on February 2024.
- The existing and new Landing Light Arrays are conditionally interchangeable (inboard and outboard arrays must be replaced together).
- The new assemblies are now available for purchase through Boeing Distribution Incorporated (BDI).
- A service bulletin was released to allow the installation of the new RTO assembly on 737 MAX airplanes prior to line number 8900. The Service Bulletin provides retrofit instructions to resolve a potential riding condition of the new RTO assembly and a fastener.
- Honeywell's investigation on the Power Supply failures found the root cause to be cracking solder joints on specific resistor installations.
- Honeywell is also investigating additional failure modes on the Power Supply related to low time failures on recently delivered airplanes. ECD for Corrective Action determination is October 2026.

Interim Action

N/A

Final Action

- Honeywell redesigned the RTO assembly and the Main Array assembly of the LTRTL system to address both discolored LED and solder joint failure modes.
- New Landing Lights part numbers: 72500454-001/-002/-003/-004.
- New Runway Turnoff Lights part numbers: 72500460-001/-002.
- Boeing released Service Bulletin (SB) 737-33-1180 to provide retrofit instructions to install the new Runway Turnoff Lights on 737 MAX airplanes prior to line number 8900.
- Honeywell Service Information Letter (SIL) D202402004285 provides the support plan for existing parts after release of the new LTRTL design.
  - The Honeywell SIL can be accessed through Honeywell's MyAerospace website, https://aerospace.honeywell.com/us/en/secure/myaerospace-home
  - SIL D202402004285 Rev 1 was released in March 2026 to now allow an Operator to request an exchange directly from Honeywell for any return that is in warranty as opposed to a scrap certificate to be used to purchase that unit from BDI.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: December 2018

Solution Selected: October 2019 (Polysulfide potting change)

Solution Selected: March 2021 (LTRTL redesign)

Change Approved: August 2021

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Preliminary Design Review: October 2021

Critical Design Review: February 2022

Qualification Testing Complete: May 2023

Production Incorporation: February 2024

Spares Availability: June 2024

Honeywell SIL: January 2025

Runway Turnoff Light Service Bulletin: February 2025

Power Supply Improvement Corrective Action Determination: October 2026 (Target)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Information Letter (SIL)	D202402004285	1	03/25/2025
Service Bulletin (SB)	737-33-1180		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Runway Turnoff Lights		Honeywell	72500415-1/-2
Landing Light Arrays		Honeywell	72500410-1/-2/-3/-4
LTRTL Power Supply		Honeywell	72605363-1
Landing Light Arrays		Honeywell	72500454-001/-002/-003/-004
Runway Turnoff Lights		Honeywell	72500460-001/-002

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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